

2.1 Classwork: Exponential Functions Review — Markscheme

1. [Maximum mark: 7]

$$f(x) = 3 \cdot 2^x - 5.$$

- (a) As $x \rightarrow -\infty$, $2^x \rightarrow 0$, so $f(x) \rightarrow -5$.

Horizontal asymptote: $y = -5$.

A1

Do not accept -5 alone: the answer is the equation of a line.

- (b) $f(0) = 3 \cdot 2^0 - 5 = 3 - 5 = -2$, so the y -intercept is $(0, -2)$.

A1

Accept -2 .

- (c) $f(4) = 3 \cdot 2^4 - 5 = 48 - 5 = 43$.

A1

- (d) Set $f(x) = 0$: $3 \cdot 2^x = 5$, so $2^x = \frac{5}{3}$.

M1

$$x = \log_2\left(\frac{5}{3}\right) = \frac{\ln(5/3)}{\ln 2} = 0.736965 \dots$$

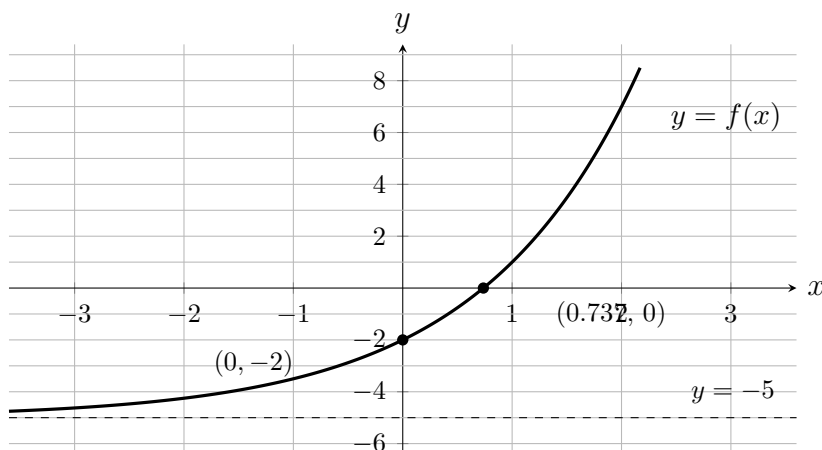
$$x \approx 0.737 \text{ (3 s.f.)}$$

A1

Accept $(0.737, 0)$. Accept the exact form $\log_2(5/3)$. A direct GDC solve of $3 \cdot 2^x - 5 = 0$ earns the M1.

- (e) Sketch:

A1A1



A1 for an increasing curve that is concave up and approaches the line $y = -5$ from above as x decreases, with that asymptote shown (dashed line, or labelled). A1 for the curve passing through both intercepts in the correct positions. A curve that cuts or crosses $y = -5$ earns neither mark.

FT: award both marks for a sketch drawn correctly from the candidate's own asymptote in (a) and intercepts in (b) and (d).

2. [Maximum mark: 6]

(a) $5^x = 300$. Take logarithms of both sides:

$$x \ln 5 = \ln 300 \quad \text{M1}$$

$$x = \frac{\ln 300}{\ln 5} = 3.543959 \dots$$

$$x \approx 3.54 \text{ (3 s.f.)} \quad \text{A1}$$

Accept $x = \log_5 300$ as the exact form. M1 is for any correct use of logarithms to bring the exponent down, in any base.

(b) $20 e^{0.15t} = 90$, so $e^{0.15t} = 4.5$.

$$0.15 t = \ln 4.5 = 1.504077 \dots \quad \text{M1}$$

$$t = \frac{\ln 4.5}{0.15} = 10.02718 \dots$$

$$t \approx 10.0 \text{ (3 s.f.)} \quad \text{A1}$$

Accept 10. A direct GDC solve of $20 e^{0.15t} = 90$ earns the M1.

(c) $2^{x+1} = 3^x$. Take logarithms of both sides:

$$(x+1) \ln 2 = x \ln 3 \quad \text{M1}$$

$$x \ln 2 + \ln 2 = x \ln 3$$

$$\ln 2 = x(\ln 3 - \ln 2)$$

$$x = \frac{\ln 2}{\ln 3 - \ln 2} = \frac{\ln 2}{\ln 1.5} = 1.709511 \dots$$

$$x \approx 1.71 \text{ (3 s.f.)} \quad \text{A1}$$

Accept the exact forms $\frac{\ln 2}{\ln 1.5}$ or $\log_{1.5} 2$. A graphical or numerical GDC solve of $2^{x+1} = 3^x$ also earns the M1.

3. [Maximum mark: 7]

$$N(t) = N_0 e^{kt}, \text{ with } N(2) = 4500 \text{ and } N(6) = 7200.$$

(a) Divide the two equations:

$$\frac{N_0 e^{6k}}{N_0 e^{2k}} = \frac{7200}{4500}, \text{ so } e^{4k} = 1.6 \quad \text{M1}$$

$$4k = \ln 1.6 = 0.470003 \dots$$

$$k = \frac{1}{4} \ln 1.6 = 0.1175009 \dots \approx 0.118 \text{ (3 s.f.)} \quad \text{A1}$$

Accept the exact form $k = \frac{1}{4} \ln 1.6$ (or $\frac{1}{4} \ln \frac{8}{5}$). A simultaneous GDC solution of the two equations for N_0 and k earns the M1.

(b) $4500 = N_0 e^{2k}$, so

$$N_0 = \frac{4500}{e^{2k}} = \frac{4500}{\sqrt{1.6}} = 3557.562 \dots \approx 3558 \quad \text{A1}$$

FT: award A1 for a value of N_0 computed correctly from the candidate's own k .

(c) One year multiplies N by e^k :

$$e^{0.1175009 \dots} - 1 = 0.1246826 \dots$$

Growth rate $\approx 12.5\%$ per year. A1

Accept $1.6^{1/4} - 1$, 12.5% , or 0.125 presented as a percentage. FT: award A1 for the percentage computed correctly from the candidate's own k .

(d) Start of 2030 is $t = 10$:

$$N(10) = N_0 e^{10k} = 4500 \times 1.6^2 = 11\,520 \quad \text{A1}$$

FT: award A1 for a value computed correctly from the candidate's own N_0 and k .

(e) Set $N_0 e^{kt} = 15\,000$:

$$e^{kt} = \frac{15\,000}{3557.562 \dots}, \text{ so } t = \frac{1}{k} \ln \left(\frac{15\,000}{N_0} \right) \quad \text{M1}$$

$$t = 12.24649 \dots \approx 12.2 \text{ years (3 s.f.)} \quad \text{A1}$$

A direct GDC solve of $N_0 e^{kt} = 15\,000$ earns the M1. Accept any value consistent with the candidate's own correctly-rounded intermediates — including 12.2 from full precision and 12.2 from $k = 0.118$, $N_0 = 3558$. Award full marks when the method is correct and the value follows from the candidate's earlier rounding.

Feedback: carry full precision through every intermediate step and round only the final answer.

4. [Maximum mark: 6]

$$FV = PV \left(1 + \frac{r}{100}\right)^n, \text{ with } PV = 2000 \text{ and } r = 4.5.$$

(a) $FV = 2000(1.045)^8 = 2844.2012\dots$

$$= \$2844.20$$

A1

(b) Require $2000(1.045)^n \geq 4000$, i.e. $(1.045)^n \geq 2$.

M1

$$n \geq \frac{\ln 2}{\ln 1.045} = 15.7473\dots, \text{ so the smallest whole number of years is } n = 16. \quad \mathbf{A1}$$

The A1 is for the integer 16; 15.7 alone is not the answer to the question. Accept a table search that shows both adjacent rows: $2000(1.045)^{15} = \$3870.56$ (under \$4000) and $2000(1.045)^{16} = \$4044.74$ (over). A single-row table still earns the mark, with feedback that the adjacent row is what justifies the crossover.

(c) Require $2000(1.045)^n > 3000(1.02)^n$.

M1

$$\left(\frac{1.045}{1.02}\right)^n > \frac{3000}{2000} = 1.5$$

$$n \ln\left(\frac{1.045}{1.02}\right) > \ln 1.5$$

$$n > \frac{\ln 1.5}{\ln(1.045/1.02)} = \frac{0.405465\dots}{0.024214\dots} = 16.7449\dots \quad \mathbf{A1}$$

Smallest whole number of years: $n = 17$.

A1

Accept a table search in place of the logarithmic method, provided both adjacent rows are shown: at $n = 16$, $\$4044.74 < \4118.36 ; at $n = 17$, $\$4226.75 > \4200.72 . A single-row table still earns the marks, with feedback that the adjacent row is what justifies the crossover.